

Cross-Domain Hybridization Engine – Full Replit Prompt with UI

Build a “Cross-Domain Hybridization Engine (CDHE)” that allows users to combine two different physical or hardware systems and generate a new hybrid system with emergent properties. CORE CONCEPT: The system converts systems into a shared representation, blends them mathematically, and outputs a new hybrid system that inherits and transforms behaviors.

----- UI SPECIFICATION (EXACT): 1. Home Screen - Centered title: “Create a New Hybrid System” - Two large side-by-side cards: - “Select System A” - “Select System B” - Below: single button “Generate Hybrid” - Minimal design, lots of whitespace 2. System Selection Modal - Grid of cards: - Name - One-line description - Small preview (text-based) - “Select” button - Top search bar 3. Loading / Thinking State - Full screen overlay: - “Analyzing systems...” - “Converting to shared representation...” - “Blending structures...” - “Generating hybrids...” - Animated progress text 4. Results Screen (Core Experience) - Display 3–5 Hybrid Cards Each Hybrid Card includes: - Title: “Hybrid #1” - Description (1–2 lines) - Sections: - “From System A” - “From System B” - “New Emergent Behavior” - Novelty Score (e.g., 87% unique) - Buttons: - “Remix” - “Save” 5. Hybrid Detail View - Expanded explanation - Simplified structure (text/diagram placeholder) - Behavior summary ----- CORE FEATURES: - System representation (graph-based) - Hybridization engine (merge + interpolate) - Emergent behavior detection - Novelty scoring - Multiple hybrid generation ----- TECH STACK (REPLIT FRIENDLY): Frontend: - React (simple components) - Minimal CSS Backend: - Python Flask - Endpoints: - /systems - /hybridize - /results Storage: - JSON or SQLite ----- LOGIC: - Convert systems to graph structures - Merge graphs - Interpolate parameters - Generate 3–5 hybrid outputs - Analyze differences to detect emergent traits ----- UX PRINCIPLES: - 3 steps only: select → select → generate - No technical input required - Fast feedback (<5–10 seconds) - Clean, minimal UI ----- GOAL: Create a system that feels like a scientific discovery tool, where users generate entirely new hybrid systems from existing ones and observe emergent behaviors that are not intuitive.